

Optimization and Curve Sketching

ANSWER KEY



Critical points and the first/second derivative tests

- Find the critical points of $f(x) = x^3 - 6x^2 + 9x$, and classify each as a local max or min using the second derivative test.
 $f'(x) = 3x^2 - 12x + 9 = 3(x-1)(x-3)$, critical points $x = 1, 3$. $f''(x) = 6x - 12$: $f''(1) = -6 < 0$ (local max); $f''(3) = 6 > 0$ (local min).
- For $f(x) = x^4 - 4x^3$, find the intervals of concavity and any inflection points.
 $f''(x) = 12x^2 - 24x = 12x(x-2)$. $f'' > 0$ (concave up) on $(-\infty, 0) \cup (2, \infty)$; $f'' < 0$ (concave down) on $(0, 2)$. Inflection points at $x = 0$ and $x = 2$.

Optimization applications

- A farmer has 200 m of fencing to enclose a rectangular field against an existing wall (so only 3 sides need fencing). Find the dimensions that maximize the enclosed area.
Let x = width (two sides), y = length (one side): $2x + y = 200$, so $y = 200 - 2x$. Area $A = xy = x(200 - 2x) = 200x - 2x^2$. $A' = 200 - 4x = 0$ gives $x = 50$, $y = 100$. Maximum area 5000 m² at width 50 m, length 100 m.
- An open-top box is made from a 12 cm square sheet by cutting equal squares of side x from each corner and folding up the sides. Find the value of x that maximizes the box's volume.
 $V(x) = x(12 - 2x)^2$, for $0 < x < 6$. $V'(x) = (12 - 2x)^2 + x \cdot 2(12 - 2x)(-2) = (12 - 2x)(12 - 6x)$.
Setting $V' = 0$: $x = 6$ (rejected, gives $V = 0$) or $x = 2$. Maximum volume at $x = 2$ cm.

More optimization

- A cylindrical can must have volume 500π cm³. Using $V = \pi r^2 h$ and surface area $S = 2\pi r^2 + 2\pi rh$, find the radius that minimizes the material used (surface area).
From V : $h = \frac{500}{r^2}$. $S(r) = 2\pi r^2 + 2\pi r \cdot \frac{500}{r^2} = 2\pi r^2 + \frac{1000\pi}{r}$. $S'(r) = 4\pi r - \frac{1000\pi}{r^2} = 0$: $4r^3 = 1000$, $r^3 = 250$, $r = \sqrt[3]{250} \approx 6.30$ cm.
- Find two positive numbers whose sum is 20 and whose product is as large as possible.
Let the numbers be x and $20 - x$: $P(x) = x(20 - x) = 20x - x^2$. $P'(x) = 20 - 2x = 0$ gives $x = 10$. Both numbers are 10, maximum product 100.

Curve sketching

- 7 For $f(x) = \frac{1}{3}x^3 - x^2 - 3x + 2$, find the intervals of increase and decrease.
 $f'(x) = x^2 - 2x - 3 = (x - 3)(x + 1)$. Critical points $x = -1, 3$. Testing: increasing on $(-\infty, -1)$, decreasing on $(-1, 3)$, increasing on $(3, \infty)$.
- 8 Using the results above, classify $x = -1$ and $x = 3$ as local maxima or minima via the first derivative test.
 f changes from increasing to decreasing at $x = -1$: local maximum. f changes from decreasing to increasing at $x = 3$: local minimum.

Minimizing distance and cost

- 9 Find the point on the parabola $y = x^2$ closest to the point $(0, 3)$, by minimizing $D^2 = x^2 + (x^2 - 3)^2$.
 $D^2(x) = x^2 + (x^2 - 3)^2$. $\frac{d}{dx}D^2 = 2x + 2(x^2 - 3)(2x) = 2x[1 + 2(x^2 - 3)] = 2x(2x^2 - 5)$. Setting to 0: $x = 0$ or $x^2 = \frac{5}{2}$. Testing shows $x = 0$ gives a local max (distance 3); $x = \pm \sqrt{5/2}$ gives the minimum, at point $(\sqrt{5/2}, \frac{5}{2})$.
- 10 A rectangular box with a square base and open top must have volume 32 m^3 . Find the base side length that minimizes the surface area, using $S = x^2 + \frac{128}{x}$ (derived from $V = x^2h = 32$).
 $S'(x) = 2x - \frac{128}{x^2} = 0$: $2x^3 = 128$, $x^3 = 64$, $x = 4 \text{ m}$.

Sketching using derivative information

- 11 For $f(x) = 3x^4 - 4x^3$, find all critical points and classify each using the second derivative test.
 $f'(x) = 12x^3 - 12x^2 = 12x^2(x - 1)$: critical points $x = 0, 1$. $f''(x) = 36x^2 - 24x$: $f''(0) = 0$ (test inconclusive, needs first derivative test — f' doesn't change sign at $x = 0$, so it's neither a max nor min); $f''(1) = 36 - 24 = 12 > 0$: local minimum at $x = 1$.
- 12 For $f(x) = xe^{-x}$, find the critical point and determine whether it is a maximum or minimum.
 $f'(x) = e^{-x} - xe^{-x} = e^{-x}(1 - x) = 0$ gives $x = 1$. $f''(x) = -e^{-x}(1 - x) - e^{-x} = e^{-x}(x - 2)$;
 $f''(1) = e^{-1}(-1) < 0$: local maximum at $x = 1$.

Visualizing critical points on a graph

- 13 The graph shows $f(x) = x^3 - 6x^2 + 9x$ from the first problem. Use the graph to confirm the local max at $x = 1$ and local min at $x = 3$ found algebraically.
The graph rises to a peak at $x = 1$ (local maximum, $f(1) = 4$), then falls to a valley at $x = 3$ (local minimum, $f(3) = 0$), matching the second-derivative-test classification.

Optimization with a fixed perimeter

- 14** A rectangular garden must have a perimeter of 80 m. Find the dimensions that maximize its area.
 $2x + 2y = 80$, so $y = 40 - x$. $A(x) = x(40 - x) = 40x - x^2$. $A'(x) = 40 - 2x = 0$ gives $x = 20$, $y = 20$.
 The maximum-area rectangle is a 20 x 20 square, area 400 m².
- 15** A Norman window (a rectangle topped by a semicircle) has perimeter 10 m, with rectangle width x and height y : $2y + x + \pi\frac{x}{2} = 10$. Express the total area $A = xy + \frac{\pi}{8}x^2$ in terms of x alone, and state the equation you would set $A'(x) = 0$ to solve (do not solve).
 From the constraint: $y = \frac{10 - x - \frac{\pi x}{2}}{2} = 5 - \frac{x}{2} - \frac{\pi x}{4}$. Substituting:
 $A(x) = x(5 - \frac{x}{2} - \frac{\pi x}{4}) + \frac{\pi}{8}x^2 = 5x - \frac{x^2}{2} - \frac{\pi x^2}{4} + \frac{\pi x^2}{8}$. Setting $A'(x) = 5 - x - \frac{\pi x}{2} + \frac{\pi x}{4} = 0$ would give the optimal x .

Concavity and the second derivative test in context

- 16** A company's profit is modeled by $P(x) = -2x^2 + 80x - 300$, where x is units sold (hundreds). Find the production level that maximizes profit, and verify it is a maximum using the second derivative.
 $P'(x) = -4x + 80 = 0$ gives $x = 20$. $P''(x) = -4 < 0$ for all x , confirming a maximum at $x = 20$ (2000 units), with maximum profit $P(20) = -800 + 1600 - 300 = 500$.